

SIFT ALGORITHM :

CONVOLUTIONS : Same thing as CNN's .
↓ this .

Gaussian blur :

2D gaussian blur :

$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-\left(\frac{x^2+y^2}{2\sigma^2}\right)} .$$

$$\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} * \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 5 & 7 \\ 11 & 13 \end{bmatrix}$$

weight kernel Feature Map .

Convolutions :

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y)$$

Blurred image at scale σ .

Small $\sigma \rightarrow$ fine details .

Large $\sigma \rightarrow$ only big structures remain .

$\hookrightarrow$ locally weighted averaging

Gaussian $\rightarrow$ blur . $\Rightarrow$ Special props :

Sobel $\rightarrow$ edges .

Laplacian $\rightarrow$ blobs .

1) Smooth continuous

2) No artifacts .

3) Separable :

$$G(x, y) = G(x) G(y)$$

Scale space :

$$L(x, y, \sigma_1), L(x, y, \sigma_2), L(x, y, \sigma_3) \dots$$

$$\sigma_{in} = k \sigma_i$$

$$k = 2^{1/5}$$

Coin on table

close $\rightarrow$ Lot of edges

far $\rightarrow$ just a blob .

$\Rightarrow$ Stack of images more blurred than prev .

SIFT $\Rightarrow$ Resize & blur .

↓
downsample
↓
efficiency

↓
simulate observing
at different scales

Octaves :

After some blurs :

$\hookrightarrow$ we downsample the image by 2 .

$\hookrightarrow$ Start again .

$\Rightarrow$

$$\sigma_i = \sigma_0 k^i \Rightarrow k = 2^{1/5}$$

$\hookrightarrow$ Smooth transition .

So: Same object at different sizes .

We capture scale invariance .

Some code stuff :

linspace (start, end, num points) .

X, Y : mesh grid () .

$$\begin{bmatrix} \text{---} \end{bmatrix} \Rightarrow X = \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}, \quad Y = \begin{bmatrix} | & | & | \\ | & | & | \\ | & | & | \end{bmatrix}$$

Convolution padding :

$$k = 3 \Rightarrow \text{pad} = k // 2 \quad \text{Image} = (h+2, k+2)$$

$$\Rightarrow \boxed{\text{pad} = 1}$$

↓
with zeros
boundary .

↓
(convolve) .

DIFFERENCE OF GAUSSIANS (DOG)

DOG $\approx$ Laplacian of Gaussian
 $= \sigma^2 \nabla^2 G$

$$L(x, y, \sigma)$$

$$D(x, y, \sigma) = L(x, y, \sigma_i) - L(x, y, \sigma_{in})$$

$\hookrightarrow$ highlights the difference .

$\hookrightarrow$ any major change pops out [blobs, corner, interest]

$\hookrightarrow$ Flat regions negate .

KEYPOINT DETECTION: FIND EXTREMA IN SCALE SPACE.

A point is a local maximum or minimum

Compare 26 neighbours:

Condition:

8 $\rightarrow$ in same image.

$D(x, y, \sigma) >$ all neighbours

9 $\rightarrow$ in scale above

$D(x, y, \sigma) <$ all neighbours

9 $\rightarrow$ in scale below

Keypoint Localization:

$$|D(x, y, \sigma)| < \text{threshold}.$$

threshold = 0.03 (typical).

Remove Edge Responses

We want corners/blobs, not edges

Hessian Matrix, $H = \begin{bmatrix} D_{xx} & D_{xy} \\ D_{xy} & D_{yy} \end{bmatrix} \Rightarrow$ $D_{xx} \rightarrow$ curvature in x direction
 $D_{xy} \rightarrow$ how x & y interact

So eigen values; $\lambda_1 \gg \lambda_2$

$D_{yy} \rightarrow$ curvature in y dir.

Instead of eigen values, SIFT uses:

$$[Tr(H)]^2 / \det(H) < (\tau+1)^2 / \tau.$$

If ratio is large $\rightarrow$ its a edge $\rightarrow$ reject.

$$Tr(H) = \lambda_1 + \lambda_2$$

$$\det(H) = \lambda_1 \lambda_2$$

$$\Rightarrow \frac{[Tr(H)]^2}{\det(H)} = \frac{(\lambda_1 + \lambda_2)^2}{\lambda_1 \lambda_2} \quad \tau = \lambda_1 / \lambda_2 \Rightarrow \frac{(\tau+1)^2}{\tau} > \frac{[Tr(H)]^2}{\det(H)}$$

Eigen Values λ_1, λ_2

$\hookrightarrow$ curvature along principal axes

Flat region: $\lambda_1 \approx 0, \lambda_2 \approx 0$

Edges: $\lambda_1 \gg \lambda_2$

Corner/blob: $\lambda_1 \approx \lambda_2 \gg 0 \rightarrow$ Strong curvature in both dir.

**TIME CONSUMING
FOR EACH PIXEL**

ORIENTATION ASSIGNMENT

Imagine: Detect a corner, rotate by 90° .

$\hookrightarrow$ Same point, but gradients change direction.

$\Rightarrow$ A dominant orientation to each keypoint.

$\Rightarrow$ Compute Image Gradients

$$G_x = L(x+1, y) - L(x-1, y).$$

$$\text{Magnitude: } m = \sqrt{G_x^2 + G_y^2}$$

$$G_y = L(x, y+1) - L(x, y-1).$$

$$\text{Orientation: } \theta = \tan^{-1}(G_y / G_x).$$

⇒ Local Neighbourhood Around Keypoint .

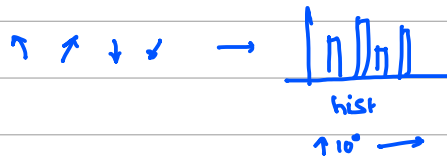
take a window around the keypoint : 16×16 .

Closer weigh more → Gaussian .

⇒ Orientation Histogram .

Build a histogram of gradient directions .

36 bins, each bin 10° , range = 0 to 360° .



DESCRIPTOR : (128 D VECTOR) .

16×16 Patch (around keypoint)

↓ Divide

$4 \times 4 \Rightarrow 16$ cells .

8 - bin histogram of orientations

$4 \times 4 \times 8 = 128 \rightarrow$ total features .